FIBRE PRODUCTS OF AFFINE SCHEMES ARE AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3405-3412, following the statement at 3398-3403.

Editorial note. The 'a fortiori' step uses the parenthetical of card L26405: the LRS fibre product of schemes is their scheme fibre product.

Allowed standard prerequisites (editorial).

- Affine schemes have finite limits computed by tensor. lemma-fibre-product-affine-schemes (card L26405, cited at ~3407). Spec(A tensor_R B) is the LRS fibre product, a fortiori the scheme fibre product.

Source reference. Original lemma 01I4: sources/stacks-schemes/pinned-original-source.tex, source lines 1202–1202. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes with the same target. If X, Y, S are all affine then $X \times_S Y$ is affine.*

Proof. Suppose that $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(R)$. By Lemma 01I4 the affine scheme $\mathrm{Spec}(A \otimes_R B)$ is the fibre product $X \times_S Y$ in the category of locally ringed spaces. Hence it is a fortiori the fibre product in the category of schemes. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.